GitHub ⭐ ⭐ ⭐ — Interview-Ready Q&A

1. What is GitHub?
GitHub is a cloud-based platform used to host Git repositories and collaborate on source code.
It provides features like repositories, branches, Pull Requests, code reviews, Issues, Actions, access control, and security features.
Teams can use GitHub as a central place to manage their application source code.
In DevOps, GitHub is commonly integrated with CI/CD tools like Jenkins and GitHub Actions to automate build, testing, and deployment.
Simple ga gurthupettuko:
Git = Version Control Tool
GitHub = Git Repository Hosting + Collaboration Platform
Eng: GitHub is a platform used to host Git repositories and provide collaboration, code review, access control, and CI/CD capabilities.

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2. What is a GitHub Repository?
A GitHub repository is a storage location for a project's source code and Git history.
It contains files, commits, branches, Pull Requests, Issues, workflows, and other project-related information. A repository can be public or private depending on the requirement.
For example, a company can maintain its application code in a private GitHub repository and allow only authorized developers to access it.
Simple ga gurthupettuko:
Repository = Project Code + Git History + Collaboration
Eng: A GitHub repository stores project source code, Git history, branches, and collaboration information.

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3. What is the difference between Public and Private Repository?
A public repository can be viewed by anyone on GitHub, while a private repository is accessible only to users who have been granted permission.
Companies normally use private repositories for internal application source code, infrastructure code, and sensitive projects.
Public repositories are commonly used for open-source projects where the source code is intentionally shared.
Simple ga gurthupettuko:
Public = Anyone can view
Private = Authorized users only
Eng: A public repository is accessible to everyone, while a private repository is restricted to authorized users.

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4. What is a GitHub Branch?
A branch is a separate line of development within a repository.
Developers normally create feature branches or bug-fix branches instead of directly modifying the main branch. Once development is completed, the changes are submitted through a Pull Request for review.
This helps multiple developers work independently without disturbing the stable code.
Simple ga gurthupettuko:
Branch = Main code ni disturb cheyyakunda separate development.
Eng: A branch provides an isolated line of development for implementing features or fixes.

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5. What is a Pull Request?
A Pull Request, or PR, is a request to merge changes from one branch into another branch.
Before merging, team members can review the code, add comments, run automated checks, and approve or reject the changes.
For example, a developer creates a feature branch, pushes the code, raises a PR to main, and the team reviews it before merging.
Simple ga gurthupettuko:
PR = "Naa changes ni review chesi target branch lo merge cheyyandi."
Eng: A Pull Request is used to review and merge changes from one branch into another.

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6. What is Code Review?
Code review is the process of checking code changes before they are merged into the main branch.
Reviewers check the code for correctness, readability, security issues, performance problems, and adherence to coding standards.
In a real project, code review also helps maintain consistent coding practices and catch defects early.
Simple ga gurthupettuko:
Code Review = Merge mundu team code ni check cheyyadam.
Eng: Code review is the process of evaluating code changes for quality, correctness, security, and maintainability before merging.

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7. What is Branch Protection?
Branch protection is a GitHub feature used to protect important branches such as main.
We can configure rules requiring Pull Requests, approvals, successful CI checks, and other conditions before changes are merged.
This prevents developers from accidentally pushing unreviewed or broken code directly into important branches.
Simple ga gurthupettuko:
Branch Protection = Main branch ki safety rules.
Eng: Branch protection enforces rules that prevent unsafe or unreviewed changes from reaching important branches.

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8. Why do we use Branch Protection Rules?
Branch protection rules help maintain the stability and quality of important branches.

For example, we can require at least one or more approvals, successful automated tests, and no unresolved review comments before merging a Pull Request.
This is especially useful in production environments where direct changes to the main branch should be controlled.
Simple ga gurthupettuko:
PR + Approval + CI Checks = Safer Main Branch
Eng: Branch protection ensures that important branches can only receive changes after required reviews and validation checks are completed.

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9. What is GitHub Actions?
GitHub Actions is a CI/CD and automation platform integrated directly with GitHub.
It can automatically build applications, run unit tests, perform security or quality checks, create artifacts, and deploy applications.
For example, a push to the main branch can trigger a workflow that builds a Docker image and deploys the application to AWS.
Simple ga gurthupettuko:
GitHub Actions = GitHub lo CI/CD automation.
Eng: GitHub Actions is a GitHub-integrated automation platform used for CI/CD workflows.

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10. What is a GitHub Actions Workflow?
A workflow is an automated process defined using a YAML file, usually stored under .github/workflows.
The workflow defines when it should run, what jobs should execute, which runner should be used, and what steps should be performed.
For example, a workflow can trigger on a Pull Request and execute build and test jobs.
Simple ga gurthupettuko:
Workflow = Complete automation definition.
Eng: A GitHub Actions workflow is a YAML-defined automation process containing triggers, jobs, and steps.

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11. What is a GitHub Actions Job?
A job is a group of related steps executed together on a runner.
A workflow can contain multiple jobs such as build, test, security scan, and deployment. Jobs can run independently or one job can depend on another.
For example:
Build → Test → Deploy
Simple ga gurthupettuko:
Workflow → Jobs → Steps
Eng: A job is a collection of steps executed together on a GitHub Actions runner.

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12. What is a GitHub Actions Runner?
A runner is the machine or execution environment where GitHub Actions jobs actually execute.
GitHub provides hosted runners, and organizations can also configure self-hosted runners when they need control over the execution environment.
For example, a self-hosted runner can be used when the pipeline needs access to internal company systems.
Simple ga gurthupettuko:
Runner = Pipeline commands execute ayye machine.
Eng: A runner is the execution environment where GitHub Actions jobs and steps are executed.

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13. What is a GitHub Actions Secret?
A GitHub Secret is a sensitive value stored securely and made available to workflows when required.
Examples include passwords, API tokens, cloud credentials, and private keys.
Instead of writing these values directly inside the source code or workflow file, we store them securely and reference them from the workflow.
Simple ga gurthupettuko:
Secret = Sensitive value ni secure ga store cheyyadam.
Eng: GitHub Secrets securely store sensitive values that workflows need without hard-coding them in source code.

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14. Why should we not store passwords or AWS credentials directly in GitHub?
Credentials stored directly in source code can accidentally become visible to developers, logs, forks, or anyone who gains access to the repository.
Even deleting the credentials later may not be sufficient because they could remain in Git history.
We should use GitHub Secrets or, where supported, short-lived authentication such as OIDC instead of long-lived credentials.
Simple ga gurthupettuko:
Credentials ni code lo hard-code cheyyakudadhu.
Eng: We should never hard-code credentials in GitHub; secure secrets or short-lived identity-based authentication should be used instead.

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15. What is a GitHub Webhook?
A webhook is a mechanism that sends an HTTP request to an external system when a specific GitHub event occurs.
For example, when code is pushed to a repository, GitHub can send a webhook notification to Jenkins, which then starts a CI/CD pipeline.
This allows GitHub repository events to integrate with external automation systems.
Simple ga gurthupettuko:
GitHub Event → Webhook → External Tool
Eng: A GitHub webhook sends event notifications to external systems so they can trigger automation.

16. How does GitHub integrate with Jenkins?
GitHub can trigger Jenkins whenever repository events such as pushes or Pull Requests occur.
A common setup uses a GitHub webhook to notify Jenkins. Jenkins then checks out the latest code and executes the configured pipeline.
For example:
Developer → GitHub → Webhook → Jenkins → Build → Test → Deploy
Simple ga gurthupettuko:
GitHub = Code/Event
Jenkins = CI/CD execution
Eng: GitHub can trigger Jenkins through webhooks, allowing Jenkins to automatically execute CI/CD pipelines after repository events.

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17. What is GitHub Fork?
A fork is a copy of another GitHub repository under your own GitHub account or organization.
Forks are commonly used when contributing to projects where you don't have direct write access to the original repository.
You can modify your fork and then create a Pull Request back to the original repository.
Simple ga gurthupettuko:
Fork = Vere repository ki mana GitHub account lo copy.
Eng: A fork creates a separate copy of a GitHub repository under another account or organization.

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18. What is the difference between Fork and Clone?
A **fork** creates a repository copy on GitHub under another account or organization.
A **clone** downloads a repository from a remote location to your local machine.
For example, in an open-source project, I may first fork the repository on GitHub and then clone my fork to my local system.
Simple ga gurthupettuko:
Fork = GitHub side copy
Clone = Local system copy
Eng: Fork creates a remote copy on GitHub, while clone creates a local copy of a repository.

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19. What is GitHub Issue?
A GitHub Issue is used to track work such as bugs, tasks, feature requests, or improvements.
Issues can be assigned to team members and organized using labels, milestones, and discussions.
For example, if a production bug is identified, the team can create an Issue, assign it to a developer, and track it until resolution.
Simple ga gurthupettuko:
Issue = Bug / Task / Feature ni track cheyyadaniki.
Eng: GitHub Issues are used to track bugs, tasks, feature requests, and other project work.

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20. What are GitHub Releases?
GitHub Releases are used to publish specific versions of a project.
They are usually associated with Git tags and can include release notes and downloadable artifacts.
For example, after completing a production release, a team can create v1.0.0 and publish it as a GitHub Release.
Simple ga gurthupettuko:
Tag + Version + Release Notes/Artifacts = Release
Eng: GitHub Releases provide a way to publish and distribute specific versions of a project.

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21. What is CODEOWNERS?
CODEOWNERS is used to define users or teams responsible for reviewing specific files or directories.
When a Pull Request changes those files, GitHub can automatically request reviews from the configured owners.
For example, security-related files can be assigned to the security team for mandatory review.
Simple ga gurthupettuko:
CODEOWNERS = Specific code ki responsible reviewers.
Eng: CODEOWNERS automatically identifies or requests reviews from the users or teams responsible for specific parts of a repository.

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22. What is a GitHub Organization?
A GitHub Organization is a shared workspace used by companies and teams to centrally manage repositories, users, teams, and permissions.
Instead of managing company repositories under individual developer accounts, organizations provide centralized administration.
Simple ga gurthupettuko:
Organization = Company/team GitHub workspace.
Eng: A GitHub Organization provides centralized management of repositories, teams, users, and access.

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23. What are GitHub Teams?
Teams are groups of users within a GitHub Organization.
They make it easier to manage permissions and collaboration for multiple users together.
For example, we can create DevOps, Developers, and QA teams and assign different repository permissions to each team.
Simple ga gurthupettuko:
Team = Organization lo users group.
Eng: GitHub Teams group users together to simplify collaboration and repository access management.

24. What is GitHub Environment?
A GitHub Environment represents a deployment environment such as development, staging, or production.
Each environment can have environment-specific secrets, variables, and protection rules. Production deployments can require manual approval before proceeding.
For example, developers may deploy automatically to Dev, while Production deployment requires approval.
Simple ga gurthupettuko:
Environment = Dev / Stage / Prod deployment configuration.
Eng: GitHub Environments provide environment-specific configuration, secrets, and deployment protection rules.

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25. What is the difference between GitHub Actions and Jenkins?
GitHub Actions is tightly integrated with GitHub and is convenient when source code is already hosted there.
Jenkins is a separate automation server with a large plugin ecosystem and can integrate with many source-control, build, testing, and deployment systems.
GitHub Actions is generally easier to start with for GitHub-centric projects, while Jenkins provides extensive customization and control.
Simple ga gurthupettuko:
GitHub Actions = GitHub-native CI/CD
Jenkins = Standalone + highly customizable CI/CD server
Eng: GitHub Actions is integrated into GitHub, while Jenkins is a standalone and highly extensible automation server.

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26. How does a typical GitHub-based CI/CD flow work?
First, a developer creates a branch and pushes the code to GitHub.
A Pull Request is created and automated checks such as build, testing, and code quality are executed. After review and approval, the code is merged into the main branch.
The merge can trigger another workflow that builds the application, creates an artifact or container image, and deploys it to the target environment.
Simple ga gurthupettuko:
Code → Branch → PR → Review → CI Checks → Merge → Build → Deploy
Eng: A typical GitHub CI/CD flow starts with a code change, followed by PR review, automated validation, merge, and deployment.

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27. How do you secure a GitHub Repository?
I would use private repositories where required, least-privilege access, branch protection, Pull Request reviews, secret management, and security scanning.
I would also prevent direct pushes to production branches and ensure sensitive credentials are never committed to the repository.
Repository and organization permissions should also be reviewed regularly.
Simple ga gurthupettuko:
Private + Least Privilege + Branch Protection + PR Review + Secrets + Security Scanning
Eng: I secure GitHub repositories using least-privilege access, branch protection, code reviews, secret management, and security controls.

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28. What happens when a developer pushes code to GitHub?
The commit is stored in the repository and GitHub processes any configured repository events.
Depending on the configuration, the push can trigger GitHub Actions workflows or send a webhook to an external CI/CD tool such as Jenkins.
The pipeline can then build, test, scan, package, and deploy the application.
Simple ga gurthupettuko:
Push → GitHub Event → Actions/Webhook → CI/CD
Eng: A GitHub push can trigger configured automation such as GitHub Actions or external CI/CD systems through webhooks.

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29. What is GitHub Actions Matrix Strategy?
A matrix strategy allows the same GitHub Actions job to run using multiple combinations of configuration values.
For example, we can test an application against multiple operating systems or programming language versions.
Instead of creating separate jobs manually, the matrix generates the required combinations automatically.
Simple ga gurthupettuko:
One Job → Multiple Versions/Environments → Testing
Eng: A matrix strategy allows a workflow job to run across multiple combinations of operating systems, versions, or configurations.

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30. Why is GitHub important in DevOps?
GitHub provides a central platform for source code management, collaboration, code review, access control, and automation.
It can integrate with CI/CD tools, cloud platforms, container tools, and security tools to create an automated software delivery process.
In a typical DevOps environment, GitHub can act as the source of truth from which CI/CD pipelines obtain application code and configuration.
Simple ga gurthupettuko:
GitHub = Code + Collaboration + Review + CI/CD + Security
Eng: GitHub is important in DevOps because it provides source control, collaboration, code review, security, and integration with CI/CD automation.